CV | Jim **Boelrijk**

Data Scientist, Evonik

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Profile

Data scientist with a PhD in AI4Science. I develop machine-learning methods for chemical and analytical workflows, with a focus on Bayesian optimization and experimental automation.
Current interests are PFNs, agentic tools and self-driving labs.

Work Experience

2024–now — **Data Scientist** · Evonik
Data-Driven Modelling group, Germany

- Help maintain and develop **BoFire**, the open-source Bayesian optimization framework behind AIChemBuddy, with partners including Bayer, BASF, Agilent and Imperial College London.
- Apply AI and Bayesian optimization to formulation and process development within Evonik.
- Develop automated LC method-optimization software used in multiple laboratories; associated patent applications pending.
- Build LLM-based tools for document digitization and summarization workflows; conduct investment scenario studies using linear programming and PyPSA.
- Build agentic tools for internal productivity.
- Contribute to projects within Evonik's Self Driving Labs initiative, recognized with the 2026 CPHI Pharma Award in Digital Transformation.

2024 — **Senior Data Science Consultant** · Valcon
Netherlands

2019–2020 — **Software Tester** · Software for Chemistry & Materials (SCM)
Netherlands

2013–2015 — **Professional Rugby Player** · Stade Montois Rugby
France

Education

2024 — **PhD Machine Learning & Chemistry**
Amsterdam Machine Learning Lab, University of Amsterdam

2019 — **Master Chemistry (Cum Laude)**
University of Amsterdam
Track: Computational Chemistry

2017 — **Bachelor Chemistry**
Vrije Universiteit Amsterdam
Minor: Programming

Selected Publications

Shape-Aware Gaussian Process Kernels for Closed-Loop Bayesian Optimization of Gradients in Automated LC Method Development
ChemRxiv, 2026 Jim Boelrijk, Sebastian Detlefsen, Johannes P. Dürholt, Christian P. Haas Preprint

Comparison of optimization algorithms for automated method development of gradient profiles
Journal of Chromatography A, 2025 Gerben B. van Henten, Jim Boelrijk, Céline Kattenberg, Tijmen S. Bos, Bernd Ensing, Patrick Forré, Bob W.J. Pirok DOI

Enhancing LC×LC separations through multi-task Bayesian optimization
Journal of Chromatography A, 2024 Jim Boelrijk, Stef R.A. Molenaar, Tijmen S. Bos, Tina A. Dahlseid, Bernd Ensing, Dwight R. Stoll, Patrick Forré, Bob W.J. Pirok DOI

Closed-loop automatic gradient design for liquid chromatography using Bayesian optimization
Analytica Chimica Acta, 2023 Jim Boelrijk, Bernd Ensing, Patrick Forré, Bob W.J. Pirok DOI

Multi-objective optimization via equivariant deep hypervolume approximation
ICLR, 2023 Jim Boelrijk, Bernd Ensing, Patrick Forré arXiv

Selected Talks & Presentations

2026	Panelist, "AI-Ready Chemistry: From Hype to Implementation", *Amsterdam Chemistry Innovation Day (ACID)*.
2026	AI-driven LC method optimization, *Agilent Lab Productivity and Automation Days*, Waldbronn.
2026	Automated AI-driven LC Method Optimization, *Future-Ready Pharma Summit*.
2025	Chair, "Making Every Measurement Count: Adaptive Optimization in Analytical Technologies", *ChemAI 2025*, Amsterdam.
2021	Focus session on machine learning for chemical discovery, *CHAINS / KNCV*.
2020	Illustere Alumni event, *University of Amsterdam*.

Teaching & Reviewing

2021–2023	Molecular Simulation (PhD winter course).
2021	Deep Learning 1 (Master AI, UvA).
2020–2021	Machine Learning 2 (Master AI, UvA).
2020	Machine Learning 1 (Master AI, UvA). Role: Teaching Assistant.
2015–2019	Teacher & Exam Trainer, Physics, *Inwijs*, Netherlands.

Supervision: 2 MSc and 1 BSc students; 1 MSc and 1 BSc graduated *cum laude*.

Conference reviewing: ICML (2024), NeurIPS (2023), AISTATS (2022, **Top 10% Reviewer**), ICLR (2022).
Journal reviewing: Journal of Chromatography A, ACS Omega.

Summer Schools & Conferences

2024	Foundation Model Winter School, *University of Amsterdam*.
2023	ICML, ICLR and AutoML Conferences.
2023	Machine Learning Theory Boot Camp, *Centrum voor Wiskunde en Informatica (CWI)*.
2021	Deep Learning & Reinforcement Learning Summer School (DLRL), *CIFAR*.

Sport Achievements

2024	Captain and Dutch national champion with RC 't Gooi.
2018, 2013	Dutch national champion with RC 't Gooi.
2014–2024	Netherlands national rugby team player.
2012	Rugby Youth Player of the Netherlands; awarded best under-18 player.

Programming

Python ●●●●●○
PyTorch ●●●●○○
GPyTorch, BoTorch, Ax ●●●●●○
NumPy, pandas, scikit-learn ●●●●○○
RDKit, ASE, CP2K, ADF ●●●●○○

Additional tools: BoFire, PyPSA. Familiarity with Slurm, Bash, Microsoft Azure and Git.

Machine Learning

Gaussian processes ●●●●○○
Bayesian optimization ●●●●●○
Multi-objective optimization ●●●●○○
(Equivariant) deep learning ●●●●○○
Molecular modeling ●●●●○○

Certificates, Languages & Interests

Certificates: Microsoft Certified: Azure AI Fundamentals (2024).
Databricks Certified Machine Learning Associate (2024–2026).
Languages: Dutch (C2), English (C2), French (B2), German (B2).
Hobbies: Weightlifting, beer brewing.